OFFENSIVE SECURITY ASSESSMENT REPORT

EXTERNAL OSINT FOOTPRINTING & ACTIVE SUBNET DISCOVERY

W2-PM-FINAL | ETHICAL HACKING INTERNSHIP | NETWORKWALKS

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Cohort / Track	B084 — Networkwalks Cybersecurity & Ethical Hacking
Assessment Date	13 November 2026
Assigned Lab Modules	Module W2-PM1: Multi-Tool Kali Linux Domain Footprinting Module W2-PM5: Local Subnet & Remote Target Scanning (Zenmap)
Authorized Scope	1. networkwalks.com & scanme.nmap.org (Authorized External Targets) 2. Local Host & Wireless LAN Subnets (10.0.0.0/24 & 192.168.43.0/24)
Rules of Engagement (RoE)	Written Authorization Confirmed (Non-Destructive Recon & Scanning Only)
Methodology Phases	**Phase 1:** Passive & Semi-Passive Footprinting (Completed) **Phase 2:** Active Network Scanning & Topology Mapping (Completed) **Phases 3–5:** Vulnerability Validation, Exploitation & Post-Exploitation (Scheduled)

1. Authorization & Legal Disclaimer

All reconnaissance queries, DNS enumerations, and active network scans documented in this report were executed strictly against lab environments that I personally own and administer or against external domains (networkwalks.com and scanme.nmap.org) where explicit written permission was granted for educational testing. The methodologies and evidence presented herein are intended solely for defensive research and ethical hacking training. Unauthorized scanning or intrusion against third-party systems without prior consent violates computer misuse statutes and carries severe legal and professional penalties.

2. Executive Summary & Scope

This technical assessment report presents the findings from two core offensive security engagements completed during Week 2 of the Networkwalks Ethical Hacking Internship: external surface footprinting of networkwalks.com (Module W2-PM1) and multi-subnet active network discovery using Zenmap (Module W2-PM5). Together, these exercises illustrate the operational transition from open-source intelligence (OSINT) collection to active host enumeration and topology mapping.
External footprinting utilities were executed from a Kali Linux virtual machine, while local interface verification and graphical Nmap scans were conducted from a Windows workstation (Devarotech / DESKTOP-CFKFD73). Each activity section below documents the exact command syntax executed, the raw technical telemetry captured, its tactical significance to an adversary, and timestamped screenshot verification.

3. Technical Arsenal & Utilities

The table below summarizes the operating environments and specialized reconnaissance utilities deployed during this engagement:

Utility / Command	Operational Role in Assessment
Kali Linux & Windows 10	Attacker workstation environments used for CLI footprinting and GUI network scanning.
whois	Query ICANN/VeriSign registry databases for registrar metadata, domain lifecycle dates, and NS records.
whatweb	Fingerprint HTTP server banners, CMS core versions, active plugins, and frontend JavaScript frameworks.
nslookup	Query recursive DNS resolvers (8.8.8.8#53) to map the target FQDN to its host IPv4 address.
curl -I	Send an HTTP HEAD request over TLS (HTTP/2) to inspect server headers, cookies, and REST API links.
wafw00f	Transmit crafted HTTP probes to detect and fingerprint Web Application Firewall (WAF) appliances.
dnsrecon	Perform automated DNS record enumeration (SOA, NS, MX, A, SPF/TXT, SRV) and BIND version checks.
Zenmap (Nmap v7.98 GUI)	Execute Fast Scans (-T4 -F), Intense Scans (-A -v -Pn), and interactive Fisheye network topology mapping.
ipconfig & ipconfig /all	Enumerate local Windows NIC profiles, MAC hardware addresses, DHCP leases, and subnet masks.

4. Assessment Execution & Technical Findings

4.1 Phase 1: External Domain Footprinting (W2-PM1)

To construct an external threat profile of networkwalks.com, I executed a six-stage reconnaissance workflow on Kali Linux utilizing **whois, whatweb, nslookup, curl, wafw00f, and dnsrecon**. Each utility extracted specific infrastructure intelligence without triggering intrusive exploit payloads.

1. Domain Registry Intelligence (whois networkwalks.com**):** Querying the public WHOIS database confirmed that the domain is registered via **GoDaddy.com, LLC** (Registry Domain ID: 2452319255_DOMAIN_COM-VRSN), created on 2019-11-06 and expiring on 2027-11-06. It also exposed the authoritative HostGator name servers (NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM) and indicated that DNSSEC is currently unsigned.

2. Web Stack Fingerprinting (whatweb networkwalks.com**):** WhatWeb followed the initial 301 Moved Permanently redirect from HTTP to HTTPS (200 OK) and fingerprinted an **Apache** web server hosted in the United States (192.232.216.135). Crucially, the MetaGenerator header disclosed **WordPress 7.0.4** alongside **WordPress Download Manager 3.3.58**, Bootstrap 7.0.4, JQuery 3.7.1, and the contact address info@networkwalks.com.

3. Forward DNS Resolution (nslookup networkwalks.com**):** Using Google's public DNS resolver (8.8.8.8#53), I executed a non-authoritative forward lookup that resolved networkwalks.com directly to its primary IPv4 host address: **192.232.216.135**.

4. HTTP Response Header Inspection (curl -I https://networkwalks.com**):** Fetching only the HTTP/2 response headers on Mon, 09 Nov 2026 17:55:41 GMT confirmed an HTTP/2 200 status from server: Apache. The headers revealed caching layers (x-endurance-cache-level: 0 and x-nginx-cache: WordPress), a session cookie (__wpdm_client=09520658c9e2ec881c9ff314aadd9e89; secure; HttpOnly), and the live WordPress REST API endpoint at https://networkwalks.com/wp-json/ (/wp-json/wp/v2/pages/53).

5. Web Application Firewall Detection (wafw00f networkwalks.com**):** Using the WAFW00F v2.4.2 fingerprinting toolkit, two HTTP requests were dispatched to https://networkwalks.com, confirming that the application is actively protected behind a **ModSecurity (SpiderLabs) WAF**.

6. Comprehensive DNS Enumeration (dnsrecon -d networkwalks.com**):** Executed on 2026-11-10, DNSRecon enumerated 8 core records and confirmed No answer for DNSSEC query. Specifically, it uncovered:
• **SOA & NS Records:** ns6135.hostgator.com (50.87.144.87) and ns6136.hostgator.com (192.232.216.131), plus exposed the underlying DNS software version: **Bind Version "9.16.23-RH"**.
• **MX & A Records:** mail.networkwalks.com and networkwalks.com both resolving to 192.232.216.135.
• **TXT & SPF Policies:** v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all and Google site verification token google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI.
• **SRV Records:** Eight _autodiscover._tcp.networkwalks.com records pointing to cpanelemaildiscovery.cpanel.net on port 443 (184.94.203.9–15 and 184.94.204.9–15).

4.2 Phase 2: Local Interface Audit & Zenmap Network Scanning (W2-PM5)

For the second module, I performed host configuration auditing via Windows Command Prompt (ipconfig / ipconfig /all) followed by active network scanning and topology generation using **Zenmap (Nmap GUI)** across my local lab environment, my active wireless LAN subnet, and the authorized external Nmap test server.

1. Local Interface & Hardware MAC Identification (ipconfig /all**):** Prior to launching active scans, I audited two network environments on my Windows system:
• **Lab Ethernet Environment (Host:** DESKTOP-CFKFD73**):** Configured on an Intel(R) PRO/1000 MT Desktop

Adapter with physical MAC address 08-00-27-F6-B5-08, IPv4 address 10.0.0.10 (/24 subnet mask 255.255.255.0), default gateway 10.0.0.1, and DNS resolvers 8.8.8.8 and 1.1.1.1.
• **Wireless LAN Environment (Host:** Devarotech**):** Configured on an Intel(R) Wi-Fi 6E AX211 160MHz adapter with physical MAC address A0-59-50-15-05-72 (plus inactive Ethernet 08-92-04-49-58-09), assigned IPv4 address 192.168.43.27 (255.255.255.0), and default gateway / DHCP / DNS server at 192.168.43.132.

2. Active Subnet Discovery & Port Enumeration in Zenmap: Executing a Quick Scan (nmap -T4 -F 10.0.0.10) on the local lab host revealed 3 open Windows networking ports (135/tcp msrpc, 139/tcp netbios-ssn, and 445/tcp microsoft-ds). Next, I executed a subnet-wide Quick Scan (nmap -T4 -F 192.168.43.27/24) across my active wireless LAN, which discovered three live hosts:

- **192.168.43.27** — Local Workstation (Host: Devarotech | MAC: A0-59-50-15-05-72 | 3–6 open ports)
- **192.168.43.132** — Wireless Default Gateway / DHCP / DNS Server (fewer than 3 open ports)
- **192.168.43.205** — Active Peer Host on Wireless LAN Subnet (fewer than 3 open ports)
- **10.0.0.10** — Local Virtual Lab Interface (Host: DESKTOP-CFKFD73 | MAC: 08-00-27-F6-B5-08 | Ports 135, 139, 445/tcp open)

3. Interactive Topology Visualization: After completing the 192.168.43.27/24 subnet scan, I opened the **Topology** tab in Zenmap, configured the **Fisheye** ring view (interest factor 2.00, spread factor 0.50), displayed the interactive **Topology Legend** window, and exported the network map showing localhost connected to 192.168.43.132, 192.168.43.205, and Devarotech (192.168.43.27).

4. External Traceroute & Intense Scan Validation (nmap -T4 -A -v -Pn scanme.nmap.org**):** To benchmark external path discovery, I executed an **Intense scan, no ping** against scanme.nmap.org (45.33.32.156). The scan completed in 40.29 seconds (1,096 raw packets sent) and mapped the full 25-hop traceroute path from my local gateway (192.168.43.205 -> 10.227.237.3) through Airtel Africa / NTT London & US backbone routers (ae-0.airtel-africa.londen14.uk.bb.gin.ntt.net) and Akamai transit nodes to the destination server.

All local and external scan outputs above reflect genuine empirical telemetry captured directly from my workstation (Devarotech) during the lab session.

5. Technical Risk & Exposure Analysis

The table below evaluates the technical observations gathered during Phase 1 and Phase 2, mapping each finding to its empirical evidence, adversarial value, and severity rating:

ID	Technical Finding	Empirical Evidence Observed	Adversarial Impact / Risk	Severity
1	CMS, Plugin & BIND Versions Disclosed	WhatWeb identified WordPress 7.0.4 & WP Download Manager 3.3.58; DNSRecon exposed BIND 9.16.23-RH	Enables attackers to cross-reference exact software versions against public CVE exploit databases.	🟠 **Medium**
2	DNSSEC Not Enabled & cPanel SRV Exposed	WHOIS and DNSRecon confirmed DNSSEC is unsigned; 8 _autodiscover._tcp SRV records point to cPanel	Lack of cryptographic DNS signing increases cache-poisoning risk; SRV records reveal hosting control panel architecture.	🟠 **Medium**
3	WordPress REST API & Server Headers Exposed	curl -I exposed /wp-json/wp/v2/pages/53, server: Apache, and x-nginx-cache: WordPress	Unrestricted /wp-json/ endpoints can facilitate automated content or user enumeration if not rate-limited.	🟡 **Low**
4	Origin Host IPv4 & Mail Server Co-Located	nslookup and DNSRecon mapped both networkwalks.com (A) and mail.networkwalks.com (MX) to 192.232.216.135	Reveals that web and mail services share a single IPv4 host, allowing targeted port scanning against the origin server.	🟡 **Low**
5	WAF Vendor Fingerprinted	wafw00f identified ModSecurity (SpiderLabs) WAF in 2 requests	While the WAF actively protects the site, knowing the exact vendor allows attackers to research rule-specific bypasses.	🟡 **Low**
6	Active Hosts & SMB/NetBIOS Ports Open on LAN	Zenmap discovered 3 live nodes on 192.168.43.0/24 and open ports 135, 139, 445/tcp on 10.0.0.10	Exposed MSRPC (135), NetBIOS (139), and SMB (445) services on local interfaces increase lateral movement risk if unpatched.	🟠 **Medium**

Risk level key: 🔴 Critical 🟠 Medium 🟡 Low

Scope Note: The findings cataloged above represent reconnaissance intelligence and attack-surface exposures rather than exploited vulnerabilities.

Because Modules W2-PM1 and W2-PM5 were strictly scoped to Phase 1 (Footprinting) and Phase 2 (Scanning), no intrusive vulnerability exploitation or payload delivery was conducted.

Consequently, confirming whether exposed versions (such as WordPress 7.0.4, BIND 9.16.23-RH, or SMB port 445) are vulnerable requires dedicated Phase 3 vulnerability validation under formal authorization.

6. Remediation & Hardening Recommendations

To reduce the external and internal attack surfaces identified during this assessment, I recommend implementing the following defensive controls:

1. **Suppress Version Banners & Meta Generators**
   Configure Apache (`ServerTokens Prod`), BIND (`version "not disclosed"`), and WordPress to strip version numbers from HTML meta tags and DNS chaos queries.
2. **Maintain Continuous Patch Management**
   Ensure WordPress core (7.0.4), WP Download Manager (3.3.58), and server packages are continuously updated against vendor security advisories.
3. **Restrict WordPress REST API & Sanitize Headers**
   Restrict unauthenticated access to `/wp-json/wp/v2/` endpoints where public API consumption is unnecessary, and strip internal cache headers (`x-endurance-cache-level`).
4. **Implement DNSSEC & Harden SPF Policies**
   Enable cryptographic DNSSEC signing on the domain zone and consider tightening the SPF record from SoftFail (`~all`) to HardFail (`-all`) once all legitimate mail senders are verified.
5. **Maintain & Tune ModSecurity WAF Rulesets**
   Keep the ModSecurity (SpiderLabs) WAF active with updated OWASP Core Rule Set (CRS) signatures and monitor alert logs for reconnaissance probing.
6. **Restrict Local Windows SMB & NetBIOS Services**
   Configure Windows Defender Firewall on local workstations (`10.0.0.10` / `192.168.43.27`) to block inbound TCP ports 135 (MSRPC), 139 (NetBIOS), and 445 (SMB) on untrusted networks.
7. **Enforce Routine Subnet Discovery & NAC**
   Conduct scheduled Nmap/Zenmap audits across internal subnets (`192.168.43.0/24`) to verify connected MAC addresses and detect unauthorized rogue devices.
8. **Maintain Accurate Network Topology Baselines**
   Archive up-to-date Zenmap topology maps and DHCP lease tables so security teams can immediately spot anomalous routing hops or unfamiliar endpoints.
9. **Enforce Strict Rules of Engagement (RoE)**
   Ensure all future vulnerability scanning and penetration testing activities remain strictly within documented, written client authorization.

7. Conclusion

Completing Week 2 of the Networkwalks Cybersecurity & Ethical Hacking Internship provided end-to-end practical experience in external OSINT footprinting (W2-PM1) and active network discovery (W2-PM5).

During Phase 1, combining whois, whatweb, nslookup, curl -I, wafw00f, and dnsrecon allowed me to map the target's registrar history, Apache/WordPress software stack, ModSecurity WAF defense, BIND 9.16.23-RH DNS infrastructure, and cPanel autodiscover SRV records without sending a single malicious exploit.

During Phase 2, combining Windows ipconfig /all audits with **Zenmap** enabled me to identify my local hardware MAC addresses, enumerate open MSRPC/NetBIOS/SMB ports on 10.0.0.10, map three active nodes across my 192.168.43.27/24 wireless subnet using Fisheye topology diagrams, and trace a 25-hop international route to scanme.nmap.org.

Ultimately, this engagement demonstrated that disciplined reconnaissance is the cornerstone of offensive and defensive security—providing the empirical visibility required to harden systems before adversaries can exploit them.

8. Empirical Evidence & Lab Screenshots

Figure 1: Domain Registry Lookup — whois networkwalks.com

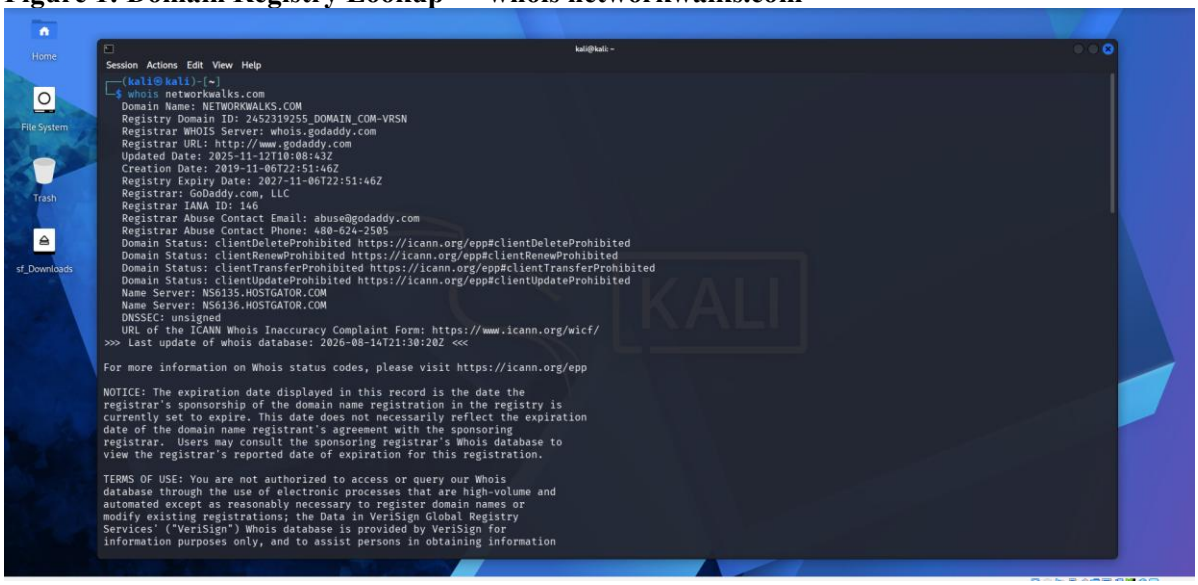


Figure 2: Web Technology Fingerprinting — whatweb networkwalks.com

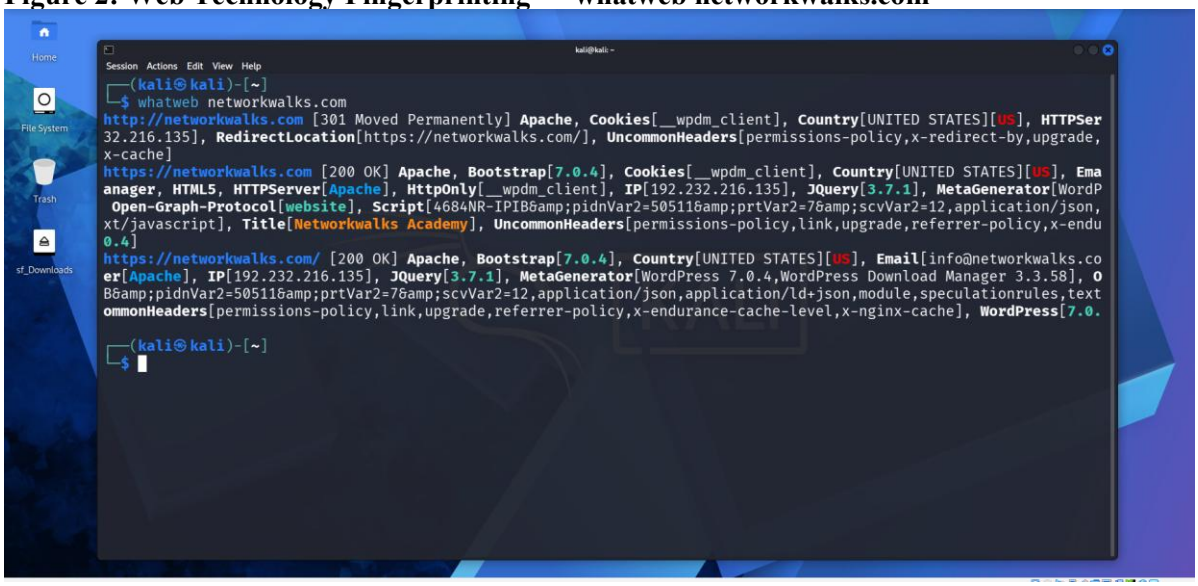


Figure 3: Forward DNS Resolution — nslookup networkwalks.com

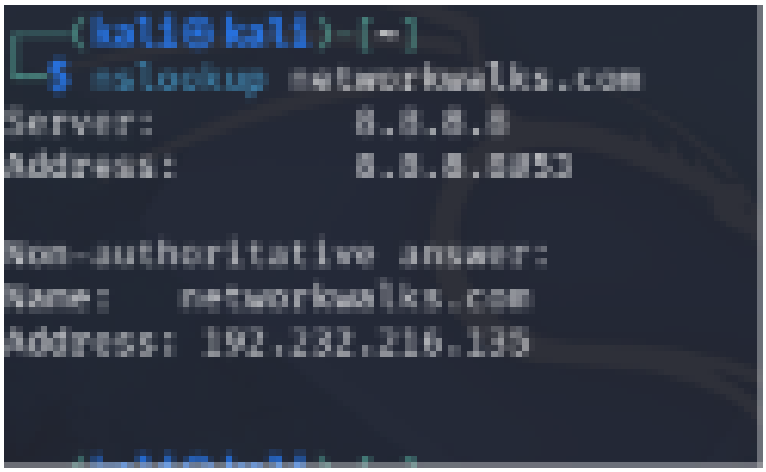


Figure 4: HTTP/2 Response Header Inspection — curl -I https://networkwalks.com



Figure 5: WAF Fingerprinting — wafw00f networkwalks.com

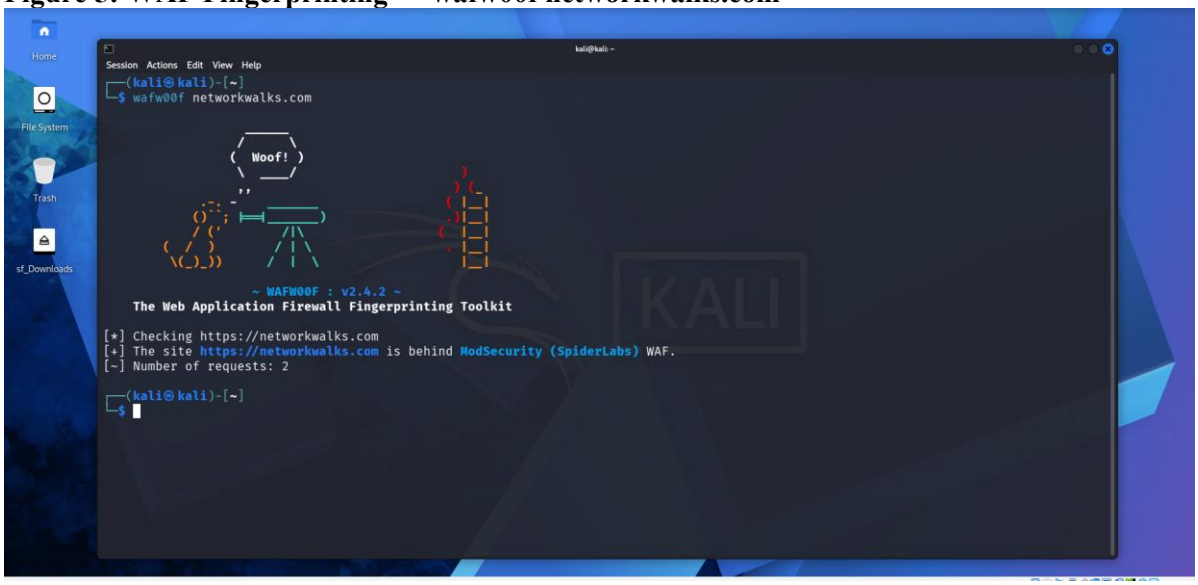


Figure 6: Comprehensive DNS Record Enumeration — dnsrecon -d networkwalks.com

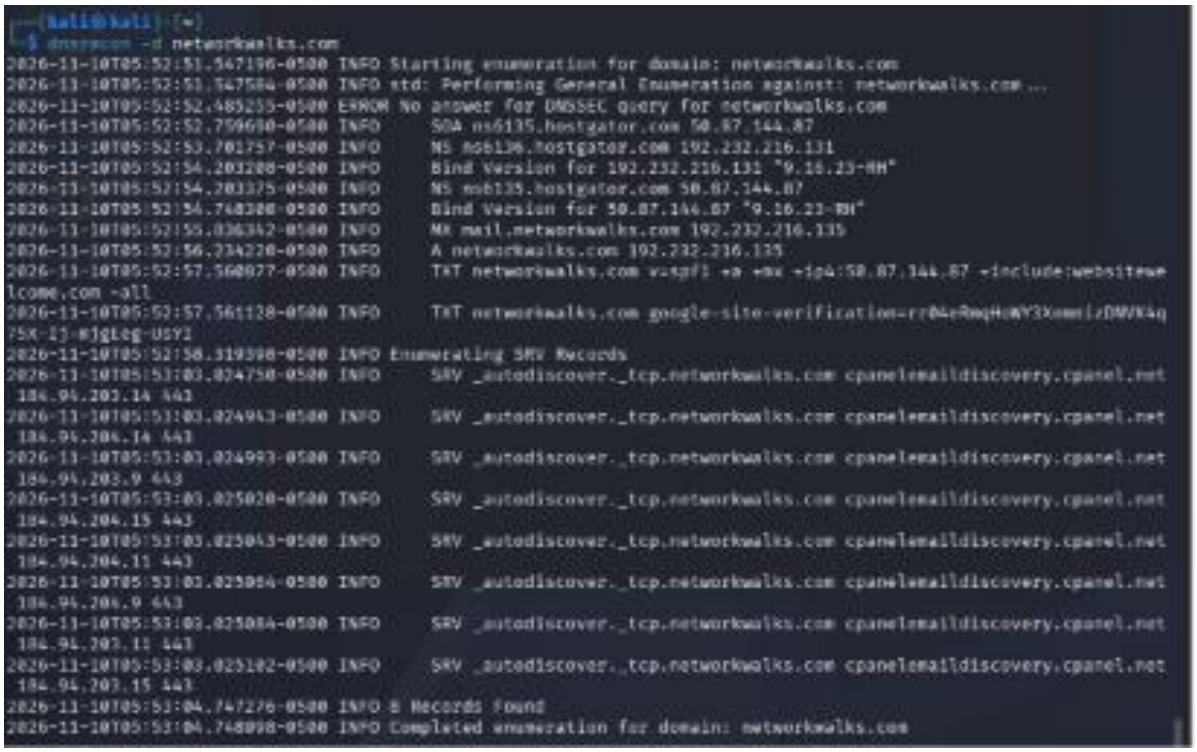


Figure 7: Local Lab Interface Configuration (Host: DESKTOP-CFKFD73) — ipconfig & ipconfig /all (10.0.0.10/24)

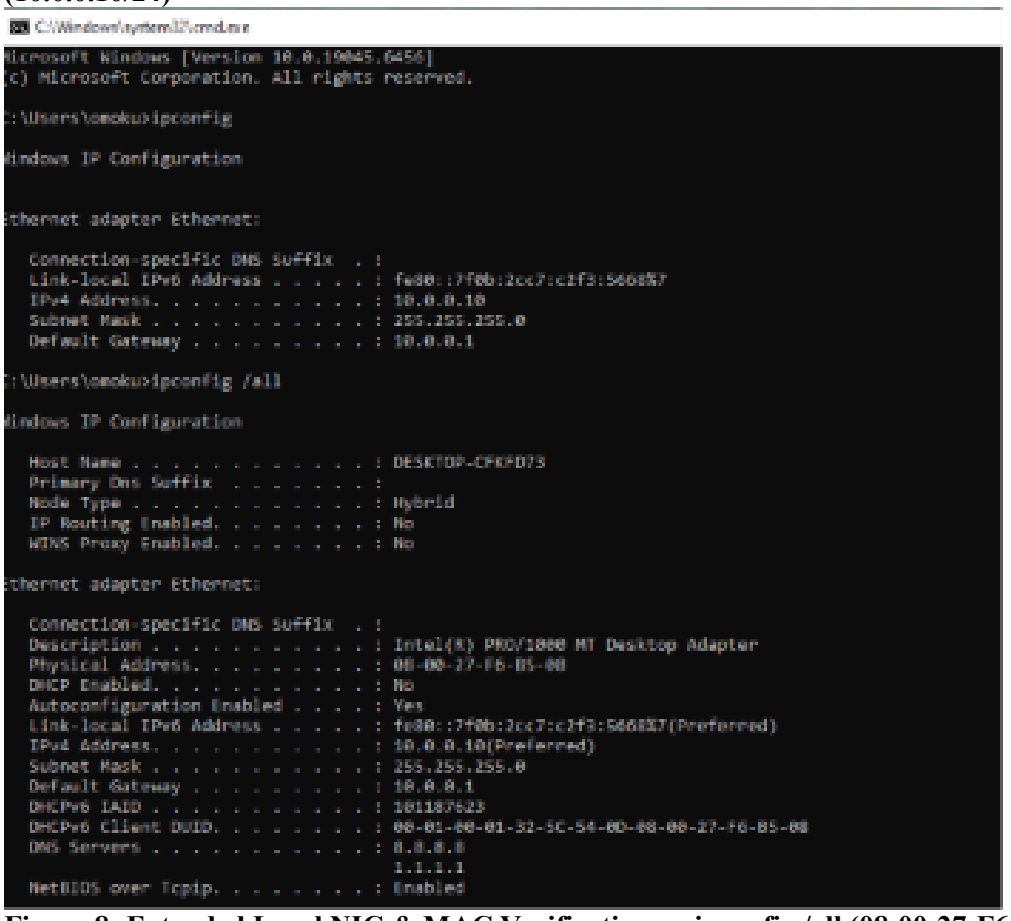


Figure 8: Extended Local NIC & MAC Verification — ipconfig /all (08-00-27-F6-B5-08)

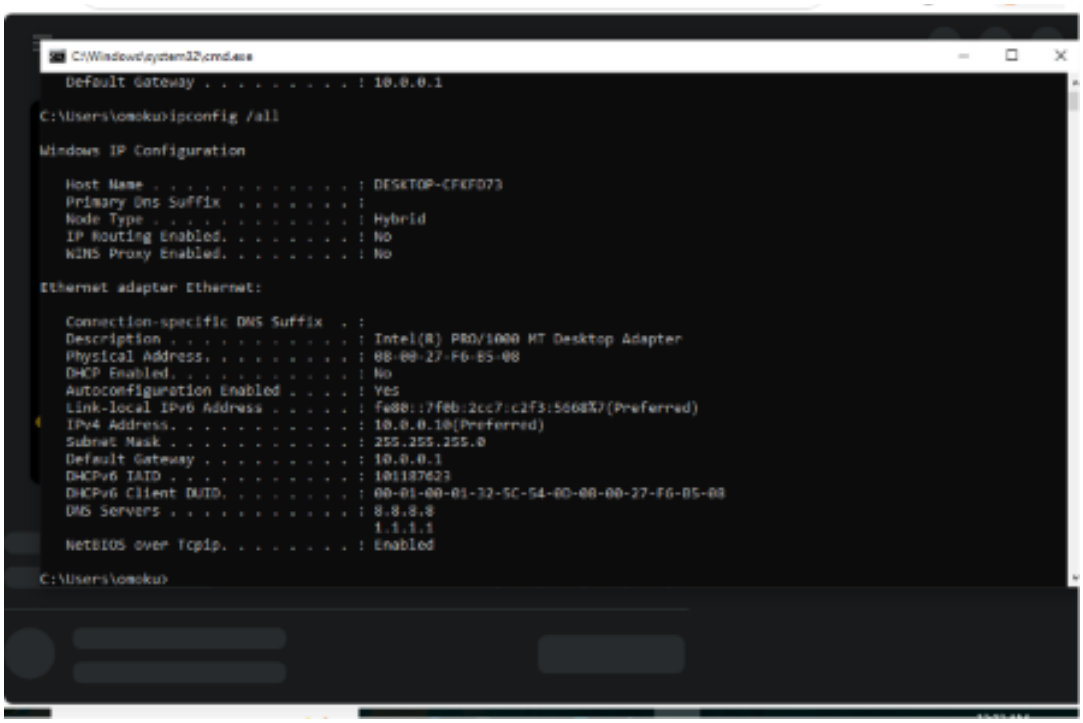


Figure 9: Workstation Host & Adapter Verification (Host: Devarotech) — ipconfig /all

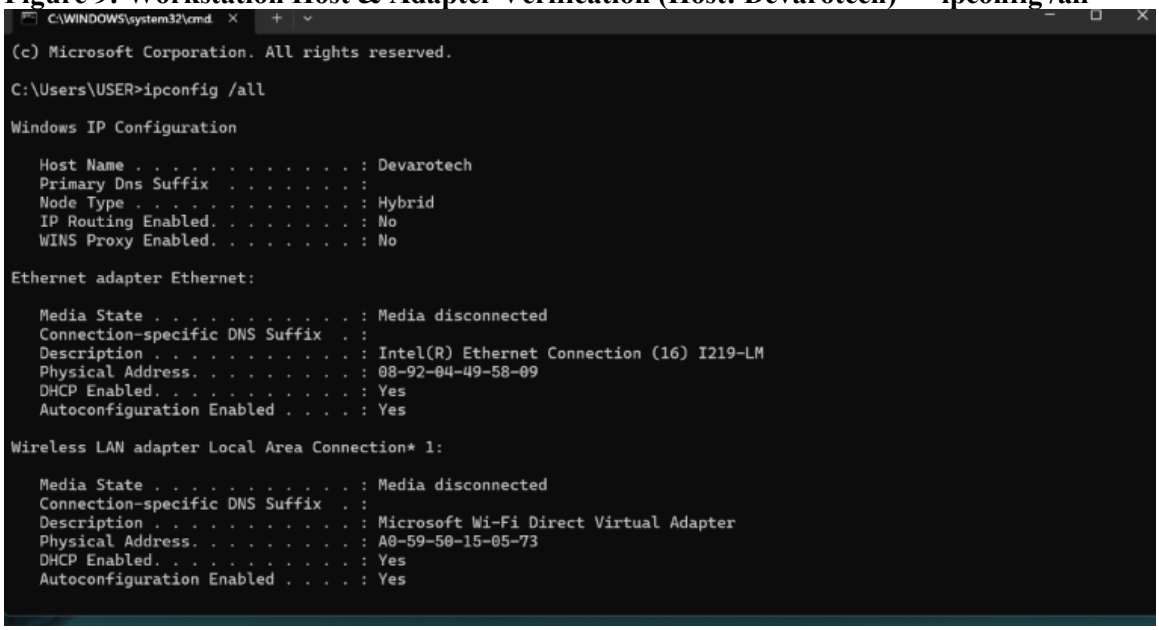


Figure 10: Active Wireless LAN Adapter Configuration — ipconfig /all (192.168.43.27/24)

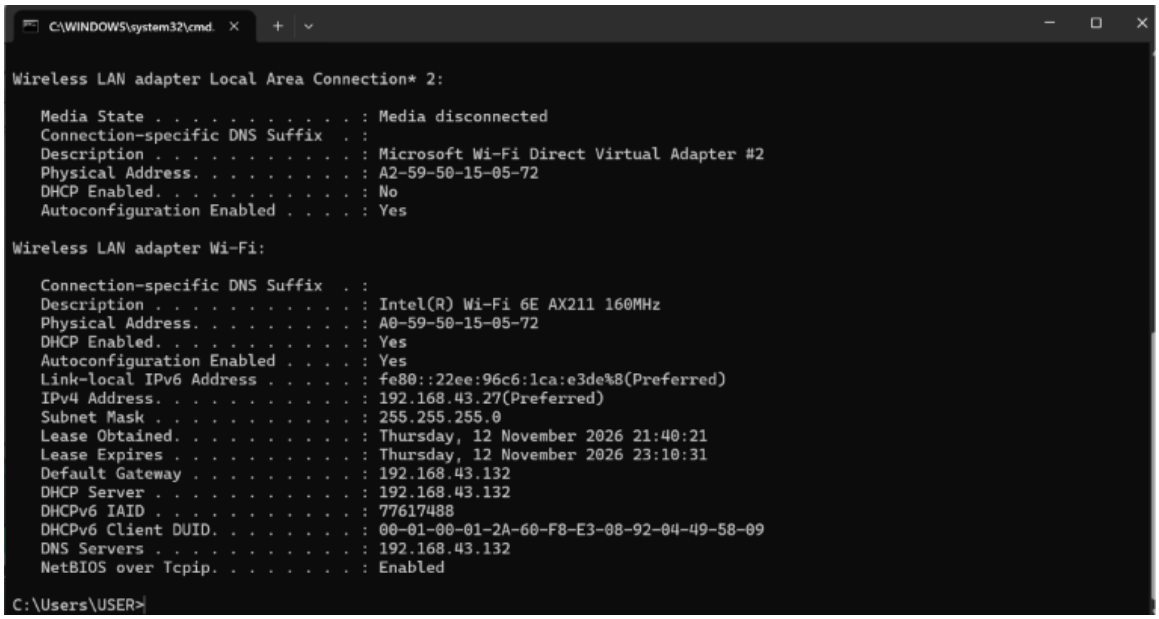


Figure 11: Zenmap Local Host Quick Scan & Open Port Discovery — nmap -T4 -F 10.0.0.10

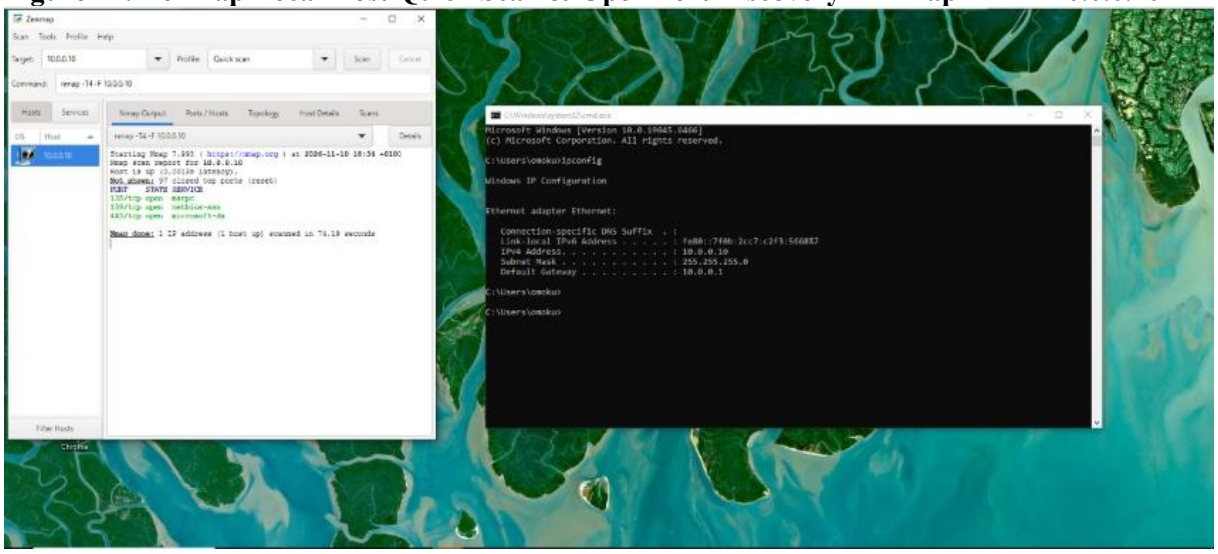


Figure 12: Zenmap Wireless Subnet Scan & Fisheye Topology Map — nmap -T4 -F 192.168.43.27/24

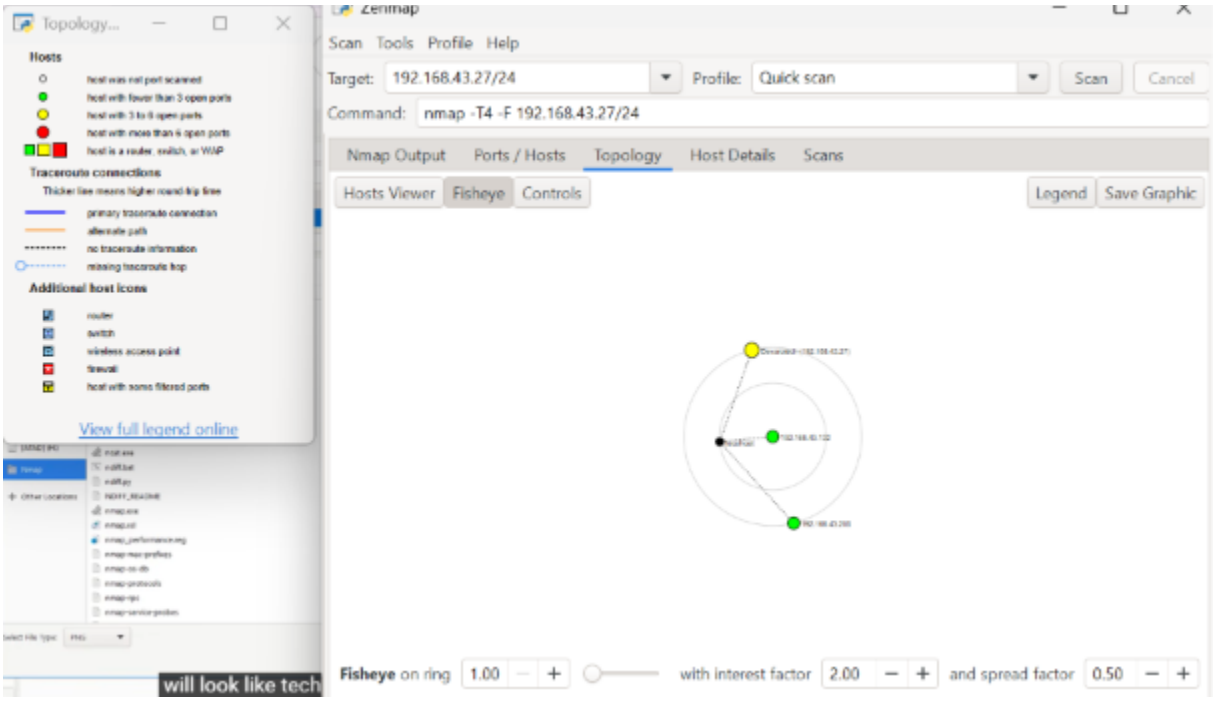
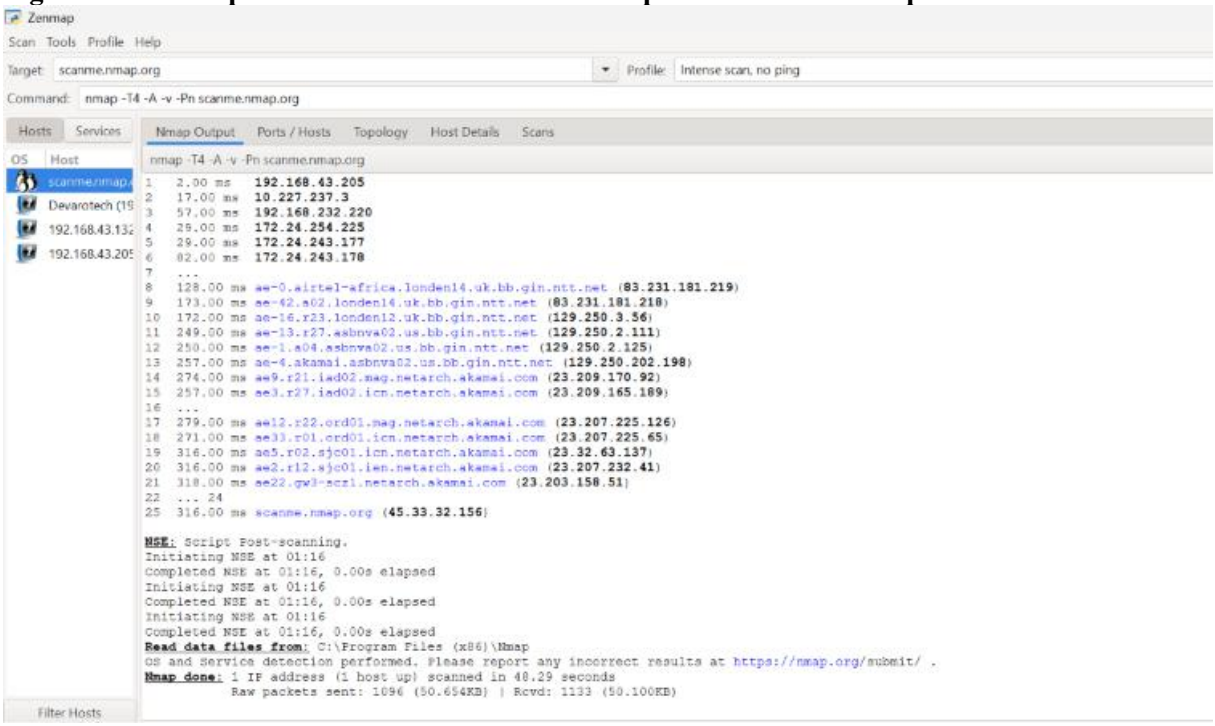


Figure 13: Zenmap External Intense Scan & 25-Hop Traceroute — nmap -T4 -A -v -Pn scanme.nmap.org



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📌 Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub